

Prüfer Code

Time limit: 1.00 s

Memory limit: 512 MB

A *Prüfer code* of a tree of n nodes is a sequence of $n - 2$ integers that uniquely specifies the structure of the tree.

The code is constructed as follows: As long as there are at least three nodes left, find a leaf with the smallest label, add the label of its only neighbor to the code, and remove the leaf from the tree.

Given a Prüfer code of a tree, your task is to construct the original tree.

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

The second line contains $n - 2$ integers: the Prüfer code.

Output

Print $n - 1$ lines describing the edges of the tree. Each line has to contain two integers a and b : there is an edge between nodes a and b . You can print the edges in any order.

Constraints

- $3 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 2 2 4	1 2 2 3 2 4 4 5